

STAINES RAJITH

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Address: London, United Kingdom

EDUCATION

Robotics and Artificial Intelligence MSc
Sep 2025 – Sep 2026

Queen Mary University of London

- **Key Modules:** Cognitive Robotics, Machine Learning and Artificial Intelligence, Medical Robotics and Surgical Techniques, Deep Learning for Data and Image Analysis

Robotics, Mechatronics and Control Engineering BEng (Hons)
Oct 2021 - Jun 2024

Loughborough University

- **Key Modules:** State Space Control, Manufacturing Automation and Control Systems, Electronics, Programming, Interfacing with Mechatronics Systems.
- **Key Projects:** Eco-Friendly Intra-Campus Monorail System Design, Gymnastics Stamp-bug and Rover, Roll-angle Measurement Device

SKILLS

- **Problem Solving** – Resolved an LED array failure in a robotics project by diagnosing wiring issues, testing components, and troubleshooting power supply faults. Implemented targeted repairs, restoring full functionality and improving system reliability.
- **Communication** – Enhanced coordination in a monorail design project by introducing structured check-ins. Improved workflow efficiency, minimized misunderstandings, and ensured project alignment through clear communication and active listening.
- **Teamwork** – Optimized team performance in a renewable energy project by delegating tasks based on expertise, streamlining development, and delivering solutions efficiently. This experience aligns with managing electrical projects and collaborating with production teams to improve systems and achieve right-first-time outcomes.
- **Adaptability and continuous learning** – Integrated new biocompatible materials into a Neuro-prosthesis design by researching industry advancements. Improved device performance by leveraging emerging technology, ensuring a more effective and innovative solution.
- **Engineering & Design Tools** – Proficient in industry-standard CAD software, including CATIA and Siemens NX, for advanced 3D modelling and product design, as well as AutoCAD for precise 2D engineering drafting. Experienced in Webots for simulating and analysing robotic systems and skilled in MATLAB (including Simulink) for numerical analysis, system modelling, and simulations to optimize engineering solutions.
- **Programming & Automation** – Skilled in Python for automation scripting and data analysis, with basic knowledge of C++. Experienced in ROS for robotic system development and simulation, as well as Arduino for embedded system prototyping. Knowledgeable in PLCs (Siemens) for industrial automation and control systems, leveraging automation to improve efficiency and reliability in engineering applications.

OTHER PROJECTS

Autonomous Drone Infrastructure Inspection System (In Progress)

- Designed a simulation environment for infrastructure inspections, engineering an autonomous drone with OpenCV for real-time image analysis to enable reliable detection of structural features.

- Improved stability by developing and tuning a PID control system, achieving optimised and consistent flight performance.
- Built a linear regression model on inspection data, allowing early forecasting of potential structural issues

WORK EXPERIENCE

First Quantum Mineral Ltd – Control Engineering Intern

Jul 2022 – Aug 2022

- Developed and implemented PLC software and designed traffic light systems that improved operational efficiency and functionality.
- Gained proficiency in CODESYS using structured text and ladder logic, accelerating project timelines and enhancing coding accuracy.

Trident College – Teaching Assistant

Jan 2021 - Mar 2021

- Provided personalised tutoring and guided lab experiments, leading to improved student performance and safer lab practices.
- Supported teachers with grading and classroom organisation, streamlining feedback processes and enhancing student engagement.

Smyths Toy Store – Retail Assistant

Nov 2024 – Dec 2024

- Delivered tailored customer support by resolving queries and assisting purchases, resulting in higher satisfaction and smoother transactions.
- Organised displays and maintained accurate inventory during peak season, ensuring efficient stock availability and professional merchandising.

CERTIFICATIONS

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| • EV Engineering Advanced Job Simulation – Forage (Ford) | • Career Essentials in Software Development – Microsoft |
| • Machine Learning at the edge - Arm | • AutoCAD 2D – CADD Centre Training Services Pvt. Ltd |
| • C++ programming – OpenEDG C++ Institute | • Python – Croma Campus Pvt. Ltd |
| • Programming with Python – OpenEDG Python Institute | • AWS Release and Deploy – edx |
| | • PLC Programming – Udemy |

EXTRACURRICULARS

Languages

- English (Native), Tamil (Native), French (Basic).

Volunteering

- Coordinated annual day trips from Trident College to Maheba Refugee Camp (UNHCR) (2018-2019).

Activities

- Cycling, Cross Country, Squash, Rugby, Cricket, Music.